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# ShopSphere Conversion Intelligence

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*Four Teams. One Unanswered Question.*



# Agenda

- |    |                                                                                     |                                                |    |                                                                                       |                                               |
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# Four Teams, One Unanswered Question

ShopSphere has increased marketing spend across every channel. Traffic is growing — but CAC is rising, ROAS is declining, and four teams are each seeing a different symptom of the same undiagnosed problem.



## One question: What truly drives conversion?

# How We Approached It

Before answering "what drives conversion," we had to define the question precisely — and be honest about how we'd know if we were wrong.



## What We're Solving For

Identify the customer, behavioral, and campaign drivers of conversion — and use them to improve targeting efficiency and marketing ROI.



## What "Solved" Looks Like

An interpretable propensity model; drivers ranked by actual contribution, not guesswork; every claim tested, not assumed.



## How We'd Know If We're Fooling Ourselves

A dual-pipeline design (Model A vs. Model B) built specifically to catch the analysis lying to us before we trust it.

# Why These Techniques

*The data's shape decided the toolkit — not the reverse*

Part of not fooling ourselves was resisting the temptation to reach for every advanced technique available. Here's what we used, and — just as importantly — what we deliberately didn't.

## USED



### Random Forest + Logistic Regression

Tabular, session-aggregated data with mixed feature types — exactly what tree ensembles handle natively



### SHAP

Required to trace every recommendation back to a specific, auditable driver



### Autoencoder

The one deep-learning tool that matches an unsupervised anomaly-detection task on numeric features

## NOT USED



### CNN

Built for spatial/image data — nothing in this dataset is visual



### RNN / LSTM

Needs per-click sequential logs; this data is pre-aggregated per session



### GAN

Built to generate synthetic samples — not this project's objective

# A Complication

*Before we could answer the question, the data asked one of its own*

83.8%

observed conversion rate in the data

vs.

1–5%

what unfiltered e-commerce  
traffic typically looks like

A gap this size isn't a rounding error. Either this data means something different than it appears to —  
or it isn't behaving like real customer traffic at all.

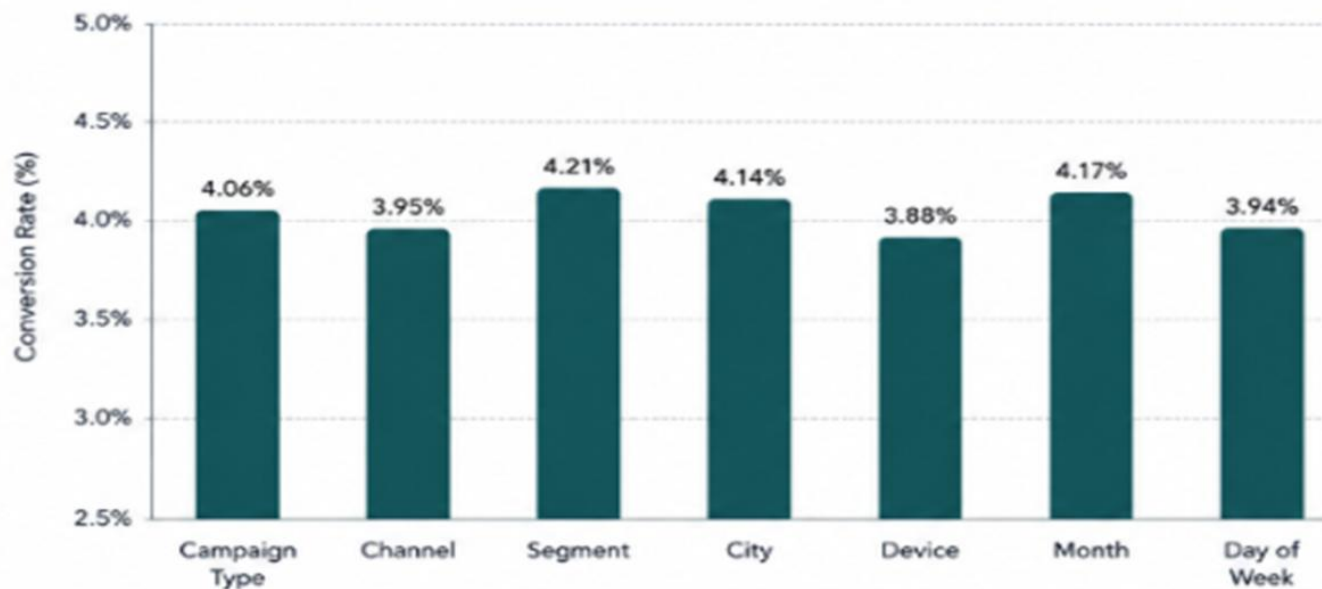
**We couldn't responsibly answer “what drives conversion” until we knew which.**

We'll return to that. For now — back to the original question: what actually drives conversion?

# Ruling Out the Obvious

*The natural suspects — channel, campaign, demographics — turned out not to be it*

Spread in Conversion Rate Across Categories



Sample sizes are well-balanced (~10,000 sessions each) across every category.



## Under 1.6 percentage points

of spread across every category tested — campaign type, channel, segment, city, device, month, day of week — with well-balanced sample sizes (~10,000 sessions each), ruling out small-sample noise as the explanation.



### First conclusion:

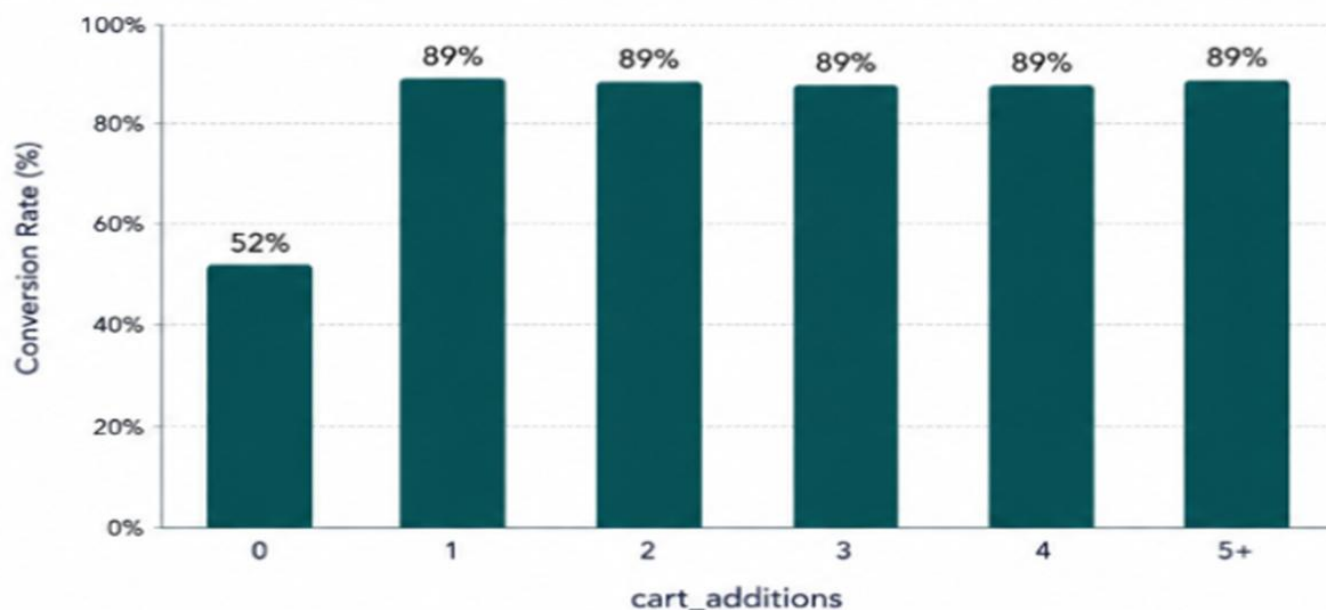
*If it's not who they are or how they arrived, it has to be something they do. (next)*

Not who they are, or how they arrived. What they **do** once they're on the site.

# The Real Driver

*In-session behavior — not acquisition — is what separates converters from everyone else*

Conversion Rate by cart\_additions (Step Function)



Pattern holds across all major channels with consistent sample sizes (~10,000 sessions each).



## A step, not a slope

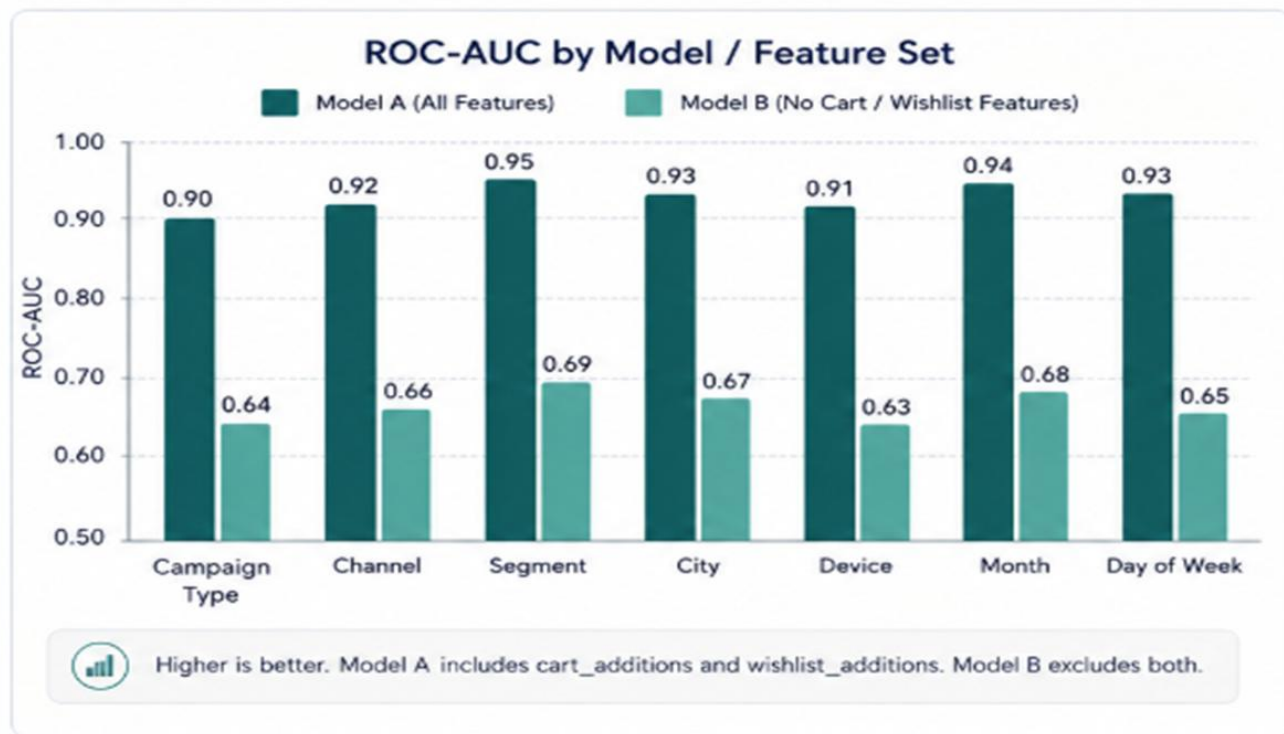
52% → 89% the moment **ANY** cart activity occurs, then flat regardless of quantity. Same shape holds — with graded severity — for wishlist\_additions and discount\_viewed.

*“But a jump this clean, this uniform across every channel, raised a different question: is this a genuine signal — or too good to be true?”*

So we built a test designed to catch ourselves being wrong.

# Stress-Testing the Finding

*Model A sees everything. Model B is deliberately blind to cart and wishlist activity. If the gap between them is small, the signal is real.*



## First, a quieter finding:

Random Forest beats Logistic Regression on every comparison — confirming the modeling approach itself holds up.



## Then, the answer we were bracing for:

The gap between Model A and Model B doesn't shrink — it grows under the simpler model. That's not what a fluke looks like. That's what a real, structural property of the data looks like.



The signal is real. But "real" and "fully understood" aren't the same thing — that reckoning comes later.

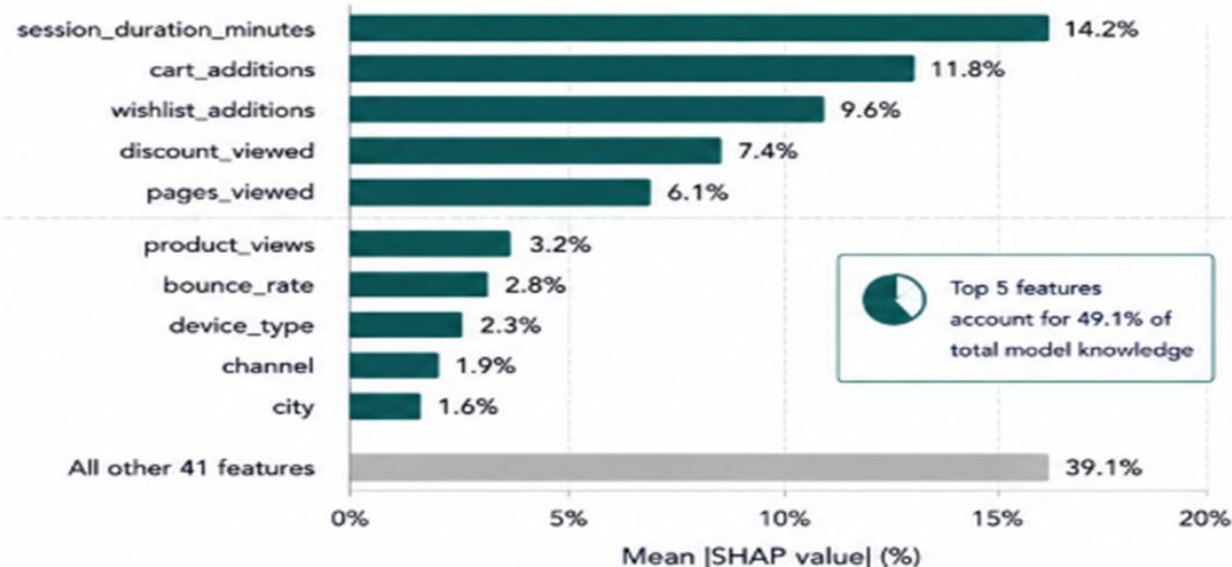
"Real" needed a number. So we asked the model to show its work.

## Precisely What Matters

*SHAP doesn't just rank features — it tells you exactly how much weight each one carries*

### SHAP Feature Importance (Random Forest)

Mean [SHAP value] (Global)



# 49.1%

of everything the model knows sits in just 5 of 51 features — cart and wishlist activity, concentrated to a degree that demands attention, not comfort.



**But the top feature wasn't cart activity at all.**

session\_duration\_minutes ranked #1 — despite showing almost no relationship with conversion in our earlier, simpler checks.



**A contradiction like that doesn't get explained away. It gets investigated.**

The contradiction had an answer: 100 sessions, all sharing one impossible number.

## An Honest Miss

*session\_duration\_minutes = 1000.0, in exactly 100 rows — a sentinel value, not real behavior.*

*So we built a detector to find things like it automatically.*

### Anomaly Detection Results

| Evaluation                                                                                                                          | What We Measured                                                             | Result                                                           | Interpretation                                          |
|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------|
|  <b>Reconstruction Error</b><br>(Isolation Forest) | Mean reconstruction error for anomalous vs. normal sessions                  | Anomalous: 0.0932<br>Normal: 0.0407<br><b>Difference: 0.0525</b> | Anomalous sessions are clearly more "abnormal"          |
|  <b>Statistical Significance</b><br>(t-test)       | Are the error distributions different?                                       | $t = 25.21$<br>$p < 0.0001$                                      | The difference is highly statistically significant      |
|  <b>Top-100 Capture</b>                            | How many of the 100 known anomalies are in the top 100 flagged by the model? | Captured: 0<br>Missed: 100<br><b>Capture Rate: 0.0%</b>          | The detector did not surface any of the known anomalies |
|  <b>Overall Discrimination</b><br>(ROC-AUC)       | How well does the model separate anomalous from normal sessions overall?     | <b>ROC-AUC: 0.6182</b>                                           | We're discrimination (only slightly better than random) |



**The test worked.  
The detector didn't.**

The difference between anomalous and normal sessions is real ( $p < 0.0001$ ) — but the detector couldn't act on it. 0 of the 100 known anomalies made the top-100 flagged list.



*We're showing you this because a deck that only shows what worked isn't showing you the whole analysis.*



**Which brings us back to the question we set aside at the start.**

Two questions have been sitting with us since the start. Time to answer both.

# Answering the Original Question



## On the 83.8%:

We can't fully resolve it — but the evidence points somewhere specific. Five separate, independent findings across this analysis (the base rate, channel uniformity, zero multicollinearity, the algorithm-independent leakage gap, and the sentinel-value outliers) all point the same direction: **this dataset behaves like it was constructed, not captured.** That's not a flaw in the analysis — it's the analysis working correctly.

- 1 Base rate: 83.8% overall conversion
- 2 Channel uniformity: <1.6 pp spread
- 3 Zero multicollinearity: no redundancy
- 4 Leakage gap persists (and widens) across models
- 5 Sentinel-value outliers: 100 sessions at 1000.0



## On what drives conversion:

Not channel. Not campaign.  
Not who someone is or how they arrived.  
**What they do in the first few minutes on the site — cart activity, wishlist activity, discount visibility — is the entire signal that matters.**



Cart Activity



Wishlist Activity



Discount Visibility



No recommendation here has been deployed.  
No before/after outcome has been measured.  
*This is diagnostic clarity — not yet a measured result.*



**But diagnostic clarity has value of its own: it stopped a plausible, expensive mistake before it happened.**

ShopSphere was set up to keep optimizing channel mix — a lever this analysis shows doesn't move the outcome. Knowing that before spending more on it is the win.



So — what should ShopSphere actually do?

In order of confidence — from what the evidence demands to what still needs proving.

# What ShopSphere Should Do Next

|    |  |                                                            |                                                                                                                                                        |                                        |
|----|--|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 01 |  | <b>Resolve the complication first</b>                      | Everything below depends on it. Validate what conversion_flag actually measures against real ShopSphere analytics before acting on anything else here. | CONFIDENCE<br>● ● ● ● ●<br>Very High   |
| 02 |  | <b>Stop optimizing channel mix</b>                         | No channel outperforms another. The lever that matters is what happens after arrival, not which channel brought the visitor.                           | CONFIDENCE<br>● ● ● ● ●<br>High        |
| 03 |  | <b>Use early-funnel signals for real-time intervention</b> | Model B — the version blind to cart activity — still predicts meaningfully. Usable today for in-session nudges before drop-off.                        | CONFIDENCE<br>● ● ● ● ●<br>Medium-High |
| 04 |  | <b>Retire click volume as a campaign KPI</b>               | ad_clicks and email_clicks show ~zero correlation with conversion.                                                                                     | CONFIDENCE<br>● ● ● ● ●<br>Medium      |
| 05 |  | <b>Treat cart/wishlist targeting as provisional</b>        | Re-run the leakage test on validated real data before committing spend to it.                                                                          | CONFIDENCE<br>● ● ● ● ●<br>Medium      |



These are the actions the evidence supports. But evidence has **boundaries** — and ours deserve to be **explicit**.



Five findings kept pointing the same direction. Here they are, together, one more time.

# Where We'd Push Further

*Not weaknesses to apologize for — the honest boundary of what this analysis can claim*



1

Conversion rate  
(83.8%) is  
**20–80× higher**  
than typical  
baselines

EVIDENCE: Slide 4



2

cart\_additions  
effect is  
**near-identical**  
across all 5  
traffic sources

EVIDENCE: Slide 5



3

**Zero mutual  
correlation**  
among all 10  
behavioral  
features

EVIDENCE: Slide 6



4

A–B model gap  
**grows,**  
**not shrinks,**  
under a simpler  
algorithm

EVIDENCE: Slide 8



5

**100 sessions**  
share an identical,  
implausible  
outlier value  
(1000.0)

EVIDENCE: Slide 11



The evidence is internally consistent.  
The uncertainty is in the data — not the analytical method.



Every technique here would apply unchanged to real ShopSphere data.  
**What needs revisiting isn't the method — it's confirming the data matches what it's supposed to represent.**



The method is ready. **The next step is simple:** validate the data — then act on what it reveals.



# References & Resources

*Everything here is verifiable — nothing in this deck asks for trust without a way to check it*



## GitHub Repository

<https://github.com/Bala-krsna/shopsphere-conversion-intelligence>

*Full notebook, cleaned code, and run history*



## Built With

pandas, scikit-learn, SHAP, TensorFlow/Keras, scipy, matplotlib

*All open-source; no proprietary tools required*



## Full Citations & Acknowledgments

Complete literature review and reference list available in the Final Technical Report, Section 12





Thank you